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Circuit substrate and electronic device

Abstract

A circuit substrate includes a base material, a first electrode, and a second electrode. The base material has an upper side and a lower side opposite to each other in a length direction. The first electrode extends and is configured on the base material along the length direction. The second electrode is configured beside the first electrode and includes a first portion and a second portion connected to each other. The first portion is configured on the base material along the length direction. The second portion is configured on the base material along a width direction and is located between the upper side and the first electrode. A cross-sectional width of the first portion becomes larger from the lower side to the upper side.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 112104587, filed on Feb. 9, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a substrate structure and an electronic device, and in particular, to a circuit substrate and an electronic device using the circuit substrate.

Description of Related Art

(3) In the circuit design of a general power module, the outline of the electrode is roughly rectangular, which means that the two opposite sides of the electrode are linear edges, and the sides of the two adjacent electrodes are disposed in parallel. When the current flows through the chips disposed in parallel, the difference between the circuit resistance of the first chip through which the current flows and the circuit resistance of the last chip through which the current flows is at least 11%. Due to the large difference in the circuit resistance between the chips, it is impossible to obtain a uniform current density and shorten the temperature difference between the chips, and the service life of the power module may be quickly diminished under long-term operation.

SUMMARY

(4) The disclosure provides a circuit substrate, which may effectively reduce the circuit resistance between subsequent parallel electronic components configured thereon.

(5) The disclosure also provides an electronic device including the above-mentioned circuit substrate, which may have a uniform current density and may shorten the temperature difference between the electronic components so as to prolong the service life.

(6) The circuit substrate of the disclosure includes a base material, a first electrode, and a second electrode. The base material has an upper side and a lower side opposite to each other in a length direction. The first electrode extends and is configured on the base material along the length direction. The second electrode is configured beside the first electrode and includes a first portion and a second portion connected to each other. The first portion is configured on the base material along the length direction. The second portion is configured on the base material along a width direction and is located between the upper side and the first electrode. A cross-sectional width of the first portion becomes larger from the lower side to the upper side.

(7) The electronic device of the disclosure includes at least one circuit substrate and multiple electronic components. The circuit substrate includes a base material, a first electrode, and a second electrode. The base material has an upper side and a lower side opposite to each other in a length direction. The first electrode extends and is configured on the base material along the length direction. The second electrode is configured beside the first electrode and includes a first portion and a second portion connected to each other. The first portion is configured on the base material along the length direction. The second portion is configured on the base material along a width direction and is located between the upper side and the first electrode. A cross-sectional width of the first portion becomes larger from the lower side to the upper side. The electronic components are configured on the circuit substrate, and the electronic components are located on the first electrode and are connected in parallel. A current is transmitted from the lower side of the base material to the second electrode through the electronic components.

(8) In an embodiment of the disclosure, the above-mentioned first portion of the second electrode has a first side and a second side opposite to each other along the width direction. The first side is located between the first electrode and the second side. The first side is a linear edge and is parallel to the first electrode.

(9) In an embodiment of the disclosure, the above-mentioned second side is an arc edge.

(10) In an embodiment of the disclosure, the above-mentioned second side is a jagged edge.

- (11) In an embodiment of the disclosure, the above-mentioned second side is an edge composed of multiple linear shapes and multiple arc shapes alternately arranged.
- (12) Based on the above, in the design of the circuit substrate of the disclosure, the cross-sectional width of the first portion of the second electrode becomes larger from the lower side of the base material to the upper side of the base material. Subsequently when the electronic components configured on the first electrode and connected in parallel are electrically connected to the second electrode by wire bonding, the design of the first portion may effectively shorten the circuit resistance between the electronic components. By reducing the circuit resistance between the electronic components, a uniform current density may be obtained and the temperature difference between the electronic components may be shortened, so the service life of the electronic device using the circuit substrate of the disclosure may be improved.
- (13) In order to make the above-mentioned features and advantages of the disclosure clearer and easier to understand, the following embodiments are given and described in details with accompanying drawings as follows.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a circuit substrate according to an embodiment of the disclosure.
- (2) FIG. 2 is a schematic diagram of a circuit substrate according to another embodiment of the disclosure.
- (3) FIG. 3 is a schematic diagram of a circuit substrate according to another embodiment of the disclosure.
- (4) FIG. 4 is a schematic diagram of an electronic device according to an embodiment of the disclosure.
- (5) FIG. 5 is a schematic diagram of an electronic device according to another embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

- (6) FIG. 1 is a schematic diagram of a circuit substrate according to an embodiment of the disclosure. Please refer to FIG. 1. In the present embodiment, a circuit substrate **100a** includes a base material **110**, a first electrode **120**, and a second electrode **130a**. The base material **110** has an upper side **111** and a lower side **113** opposite to each other in a length direction L. The first electrode **120** extends and is configured on the base material **110** along the length direction L. The second electrode **130a** is configured beside the first electrode **120** and includes a first portion **132a** and a second portion **134** connected to each other. The first portion **132a** is configured on the base material **110** along the length direction L. The second portion **134** is configured on the base material **110** along a width direction W and is located between the upper side **111** and the first electrode **120**. The cross-sectional widths SW1 and SW1' of the first portion **132a** become larger from the lower side **113** to the upper side **111**.
- (7) In detail, in the embodiment, the lower side **113** of the base material **110** may be used as the current input terminal, and the upper side **111** of the base material **110** may be used as the current output terminal. Viewed from above, the outline of the first electrode **120** is embodied as a rectangle. The first electrode **120** may be used as a configuration area for subsequent electronic components. The first portion **132a** of the second electrode **130a** and the second portion **134a** of the second electrode **130a** are seamlessly connected to form an L-like shape. It should be noted that the second electrode **130a** may be a drain electrode or a source electrode depending on the direction in which subsequent chips are placed, and there is no limitation here. The first portion **132a** of the second electrode **130a** has a first side **133** and a second side **135a** opposite to each

other along the width direction W. The first side **133** is located between the first electrode **120** and the second side **135a**. The first side **133** is embodied as a linear edge parallel to the first electrode **120**, and the second side **135a** is embodied as an arc edge. As shown in FIG. 1, the horizontal distance (i.e., in the width direction W) between the first side **133** and the second side **135a** is the cross-sectional widths SW1 and SW1'. The cross-sectional width SW1 is smaller than the cross-sectional width SW1'. That is to say, the cross-sectional width SW1 of the first portion **132a** closer to the lower side **113** is smaller than the cross-sectional width SW1' of the first portion **132a** closer to the upper side **111**. The cross-sectional width SW1' of the first portion **132a** of the second electrode **130a** becomes larger toward the upper side **111**.

(8) In addition, please refer to FIG. 1 again. The circuit substrate **100a** of the embodiment further includes an electrode **140** and an electrode **150**, which extend and are configured on the base material **110** along the length direction L. The electrode **150** is located between the electrode **140** and the first electrode **120**. Viewed from a top view, the outline of the electrode **140** and the outline of the electrode **150** are embodied as both rectangular, but are not limited thereto. It should be noted that the electrode **140** may be a drain electrode or a source electrode depending on the direction in which subsequent chips are placed, and the electrode **150** may be, for example, a gate electrode, and there is no limitation here.

(9) It must be noted here that the following embodiments continue to use the referential numbers of the components and a part of the contents of the foregoing embodiments, wherein the same referential numbers are used to denote the same or similar components, and the description of the same technical content is omitted. For the description of the omitted part, reference may be made to the foregoing embodiments, and the details are not repeated here.

(10) FIG. 2 is a schematic diagram of a circuit substrate according to another embodiment of the disclosure. Please refer to FIG. 1 and FIG. 2 at the same time. A circuit substrate **100b** of the embodiment is similar to the circuit substrate **100a** of FIG. 1. The difference between the two is that, in the embodiment, the structural design of a first portion **132b** of a second electrode **130b** is different from the structural design of the above-mentioned first portion **132a** of the second electrode **130a**.

(11) In detail, a second side **135b** of the first portion **132b** is embodied as an edge composed of multiple linear shapes **136** and multiple arc shapes **138a**, **138b**, **138c**, **138d**, **138e**, **138f**, and **138g** alternately arranged. The lengths of the arc shapes **138a**, **138b**, **138c**, **138d**, **138e**, **138f**, and **138g** gradually increase from the upper side **111** to the lower side **113** of the base material **110**, which means that the length of the arc shape **138a** is the shortest, whereas the length of the arc shape **138g** is the longest, so that cross-sectional widths SW2 and SW2' of the first portion **132a** become larger from the lower side **113** to the upper side **111**. Here, the cross-sectional widths SW2 and SW2' are exemplified by the horizontal distances between the arc shapes **138a**, **138b**, **138c**, **138d**, **138e**, **138f**, and **138g** from the first side **133** to the second side **135b**. In short, the cross-sectional width SW2 of the first portion **132b** closer to the lower side **113** is smaller than the cross-sectional width SW2' of the first portion **132b** closer to the upper side **111**. The cross-sectional width SW2' of the first portion **132b** of the second electrode **130b** becomes larger toward the upper side **111**.

(12) FIG. 3 is a schematic diagram of a circuit substrate according to another embodiment of the disclosure. Please refer to FIG. 1 and FIG. 3 at the same time. A circuit substrate **100c** of the embodiment is similar to the circuit substrate **100a** of FIG. 1. The difference between the two is that, in the embodiment, the structural design of a first portion **132c** of a second electrode **130c** is different from the structural design of the above-mentioned first portion **132a** of the second electrode **130a**.

(13) In detail, a second side **135c** of the first portion **132c** is embodied as a jagged edge. The horizontal distance (i.e., in the width direction W) between the first side **133** and the second side **135c** is the cross-sectional widths SW3 and SW3'. The cross-sectional width SW3 is smaller than the cross-sectional width SW3'. That is to say, the cross-sectional width SW3 of the first portion

132c closer to the lower side **113** is smaller than the cross-sectional width **SW3'** of the first portion **132c** closer to the upper side **111**. The cross-sectional width **SW3'** of the first portion **132c** of the second electrode **130c** becomes larger toward the upper side **111**.

(14) It should be noted that the second sides **135a**, **135b**, and **135c** of the first portions **132a**, **132b**, and **132c** illustrated above are respectively embodied as an arc shape edge, an edge composed of linear shapes and arc shapes alternately arranged, and a jagged edge, but are not limited thereto. As long as the structural design may enable the cross-sectional width of the first portions **132a**, **132b**, and **132c** of the second electrodes **130a**, **130b**, and **130c** become larger from the lower side **113** to the upper side **111** of the base material **110**, it is considered to be within a protection scope of the disclosure.

(15) FIG. 4 is a schematic diagram of an electronic device according to an embodiment of the disclosure. Please refer to FIG. 4. In the embodiment, an electronic device **10** includes, for example, the two circuit substrates **100a** of FIG. 1, multiple electronic components **30** and **50**, and multiple bonding wires **40**. The electronic components **30** and **50** are configured on the circuit substrate **100a**. The electronic components **30** are located on the first electrode **120** and are connected in parallel. The bonding wire **40** is electrically connected to the electronic component **30** and the second electrode **130a**. A current **E** is transmitted from the lower side **113** of the base material **110** to the second electrode **130a** through the electronic component **30** and the bonding wire **40**. Here, the electronic device **10** is, for example, a power module. The electronic component **30** is, for example, a diode, and the electronic component **50** is, for example, a transistor, but are not limited thereto.

(16) In detail, the current **E** may enter the circuit substrate **100a** on the right from a conductive terminal **20a**, and flow through the electronic component **50** and enter the circuit substrate **100a** on the left from the lower side **113**. At this time, the current **E** flows through the electronic components **30** connected in parallel, and enters the second electrode **130a** through the bonding wire **40**. Since the cross-sectional widths **SW1** and **SW1'** of the second electrode **130a** become larger from the lower side **113** to the upper side **111**, the circuit resistance between the electronic components **30** may be effectively reduced. According to the simulation results, the resistance value of an electronic component **30a** is, for example, $0.84\text{ m}\Omega$ 6%, the resistance value of an electronic component **30b** is, for example, $0.81\text{ m}\Omega$ 2%, and the resistance value of an electronic component **30c** is, for example, $0.79\text{ m}\Omega$, which means that the difference in the circuit resistance between the electronic components **30** connected in parallel may be within 6%. Then, the current **E** may flow through the electronic component **50** and enter the circuit substrate **100a** on the right again from the lower side **113**, or flow from the upper side **111** of the circuit substrate **100a** on the left to two conductive terminals **20b**. The current **E** entering the circuit substrate **100a** on the right from the lower side **113** of the base material **110** flows through the electronic components **30** connected in parallel, and enters the second electrode **130a** through the bonding wire **40**. Since the cross-sectional widths **SW1** and **SW1'** of the second electrode **130a** become larger from the lower side **113** to the upper side **111**, the circuit resistance between the electronic components **30** may be effectively reduced. Due to the circuit resistance between the electronic components **30** being able to be reduced, a uniform current density may be obtained and the temperature difference between the electronic components **30** may be shortened, so the service life of the electronic device **10** of the embodiment may be improved. Finally, the current **E** may flow to a conductive terminal **20c** through the upper side **111** of the base material **110**.

(17) In another embodiment, please refer to FIG. 5. In an electronic device **10'** of the embodiment, the electronic device **10'** may also replace the bonding wire **40** of the above-mentioned embodiment by using a conductive frame **60**. That is to say, the conductive frame **60** may electrically connect the electronic component **30** and the second electrode **130a**. The current **E** is transmitted from the lower side **113** of the base material **110** to the second electrode **130a** through the electronic component **30** and the conductive frame **60**, and it is considered to be within a

protection scope of the disclosure.

(18) To sum up, in the design of the circuit substrate of the disclosure, the cross-sectional width of the first portion of the second electrode becomes larger from the lower side to the upper side of the base material. Subsequently when the electronic components configured on the first electrode and connected in parallel are electrically connected to the second electrode by wire bonding, the design of the first portion may effectively shorten the circuit resistance between the electronic components. By reducing the circuit resistance between the electronic components, a uniform current density may be obtained and the temperature difference between the electronic components may be shortened, so the service life of the electronic device using the circuit substrate of the disclosure may be improved.

(19) Although the disclosure has been described with reference to the embodiments above, the embodiments are not intended to limit the disclosure. Any person skilled in the art can make some changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the scope of the disclosure will be defined in the appended claims.

Claims

1. A circuit substrate, comprising: a base material, having an upper side and a lower side opposite to each other in a length direction; a first electrode, extending and being configured on the base material along the length direction; a second electrode, configured beside the first electrode and comprising a first portion and a second portion connected to each other, wherein the first portion is configured on the base material along the length direction, the second portion is configured on the base material along a width direction and is located between the upper side and the first electrode, and a cross-sectional width of the first portion becomes larger from the lower side to the upper side; a drain electrode or a source electrode extending and configured on the base material along the length direction; and a gate electrode extending and configured on the base material along the length direction, wherein the gate electrode is located between the drain electrode or the source electrode and the first electrode, and viewed from a top view, an outline of the drain electrode or the source electrode and an outline of the gate electrode are both rectangular.
2. The circuit substrate according to claim 1, wherein the first portion of the second electrode has a first side and a second side opposite to each other along the width direction, the first side is located between the first electrode and the second side, and the first side is a linear edge and is parallel to the first electrode.
3. The circuit substrate according to claim 2, wherein the second side is an arc edge.
4. The circuit substrate according to claim 2, wherein the second side is a jagged edge.
5. The circuit substrate according to claim 2, wherein the second side is an edge composed of a plurality of linear shapes and a plurality of arc shapes alternately arranged.
6. An electronic device, comprising: at least a circuit substrate, comprising: a base material, having an upper side and a lower side opposite to each other in a length direction; a first electrode, extending and being configured on the base material along the length direction; a second electrode, configured beside the first electrode and comprising a first portion and a second portion connected to each other, wherein the first portion is configured on the base material along the length direction, the second portion is configured on the base material along a width direction and is located between the upper side and the first electrode, and a cross-sectional width of the first portion becomes larger from the lower side to the upper side; a drain electrode or a source electrode extending and configured on the base material along the length direction; and a gate electrode extending and configured on the base material along the length direction, wherein the gate electrode is located between the drain electrode or the source electrode and the first electrode, and viewed from a top view, an outline of the drain electrode or the source electrode and an outline of the gate electrode are both rectangular; and a plurality of electronic components, configured on the

circuit substrate, wherein the electronic components are located on the first electrode and are connected in parallel, and a current is transmitted from the lower side of the base material to the second electrode through the electronic components.

7. The electronic device according to claim 6, wherein the first portion of the second electrode has a first side and a second side opposite to each other along the width direction, the first side is located between the first electrode and the second side, and the first side is a linear edge and is parallel to the first electrode.

8. The electronic device according to claim 7, wherein the second side is an arc edge.

9. The electronic device according to claim 7, wherein the second side is a jagged edge.

10. The electronic device according to claim 7, wherein the second side is an edge composed of a plurality of linear shapes and a plurality of arc shapes alternately arranged.
